University of Oklahoma
Department of Mathematics
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MATH 3333–004
Linear Algebra
Spring 2026

MIDTERM EXAM 1 (FORM A)

NAME: ______________________________

ID N°: ______________________________

INSTRUCTIONS

- This is a closed-book exam. You may not use your notes, textbook or any outside resources to assist you.
- Write your answer to each question in the space assigned to each problem. If you want the instructor to read the scratch working page for the answer to a specific problem, explicitly write that on the space assigned to said problem.
- Always remember to show your work. An answer without a proper justification will earn little to no credit, even if it is correct.
- Circle or box your final answer to each question.
- Make sure that your answers are organized and legible. Use a dark pencil or a pen. Any work that is hard or impossible to read will not be graded.
- In this exam, the use of determinants is not allowed. We will allow determinants once we cover them in this course.
- The use of calculators of any type is not allowed.

This exam consists of four problems, and your total score will be out of $10 + 10 + 10 + 10 = 40$ points.

Good Luck!

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Problem 1 (2 points per part)**.** On each of the following questions, **completely fill in** the circle next to the correct answer. Each question only has one correct answer. You do not need to justify your choices.

(1) If A is an $m \times n$ matrix, the rank r of A always satisfies:

- ○ $r \leq m$ (but not necessarily $r \leq n$).
- ○ $r \leq n$ (but not necessarily $r \leq m$).
- ○ $r \leq n$ and $r \leq m$.
- ○ None of the above.

(2) A system of linear equations is called *consistent* if:

- ○ It has exactly one solution.
- ○ It has infinitely many solutions.
- ○ It has no solutions.
- ○ It has at least one solution.

(3) If $A\mathbf{x} = \mathbf{b}$ is a *non-homogeneous* linear system and $\mathbf{x}_1$, $\mathbf{x}_2$ are solutions to it, then:

- ○ $\mathbf{x}_1 + \mathbf{x}_2$ is a solution to $A\mathbf{x} = \mathbf{b}$.
- ○ $\mathbf{x}_1 - \mathbf{x}_2$ is a solution to $A\mathbf{x} = \mathbf{b}$.
- ○ $\mathbf{x}_1 + \mathbf{x}_2$ is a solution to $A\mathbf{x} = \mathbf{0}$.
- ○ $\mathbf{x}_1 - \mathbf{x}_2$ is a solution to $A\mathbf{x} = \mathbf{0}$.

(4) Given the matrices $X = [1 \quad 0 \quad 2]$ and $Y = [2 \quad 1 \quad 3]^T$, we have:

- ○ The matrices $5X + Y$, $X(2Y)$ and $4Y^T$ exist.
- ○ The matrices XY, YX and $2X + 5Y^T$ exist.
- ○ The matrices $Y(3X^T)$, $(X + Y^T)^T$ and Y^2 exist.
- ○ The matrices X^TX, $-2X + Y$ and YX^2 exist.

(5) A homogeneous system of equations $A\mathbf{x} = \mathbf{0}$ has 20 variables and 8 equations. If the augmented matrix $[A \,|\, \mathbf{0}]$ has rank 5, then:

- ○ The system is consistent regardless of A, and its general solution involves 12 parameters.
- ○ The system is consistent regardless of A, and its general solution involves 15 parameters.
- ○ There are matrices A for which the system is inconsistent. However, when the system is consistent, its general solution involves 12 parameters.
- ○ There are matrices A for which the system is inconsistent. However, when the system is consistent, its general solution involves 15 parameters.

Problem 2 (10 points)**.** Suppose A and B are 3×2 matrices.

(a) (6 points) Find, in terms of A and B, the matrix X that satisfies

$$5\big(X + 3A^T\big)^T = 2X^T - B.$$

What is the size of X?

(b) (4 points) Calculate the matrix $C^2 - 7C + 14I_2$, where

$$C = \begin{bmatrix} 3 & -2 \\ 1 & 4 \end{bmatrix}.$$

Problem 3 (10 points)**.** Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 5 & 10 & -4 \\ 0 & 1 & 3 \end{bmatrix}.$$

(a) (6 points) Using elementary row operations, find the inverse A^{-1} of A.

(b) (4 points) Use the previous part to find the solutions to the system

$$\begin{cases} x + 2y - z = 0 \\ 5x + 10y - 4z = -1 \\ y + 3z = 3 \end{cases}$$

Problem 4 (10 points)**.** Consider the system of linear equations $A\mathbf{x} = \mathbf{b}$ whose augmented matrix is:

$$[A\,|\,\mathbf{b}] = \left[\begin{array}{ccc|c} 1 & 0 & 6 & a \\ 3 & 1 & 5 & b \\ -4 & -2 & 2 & c \end{array}\right].$$

(a) (2 points) What conditions do a, b and c have to satisfy in order for the system $A\mathbf{x} = \mathbf{b}$ to be homogeneous?

(b) (4 points) Using the Gaussian algorithm, show that the system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $c = 2a - 2b$.

(c) (4 points) Calculate a set of basic solutions to the system $A\mathbf{x} = \mathbf{0}$.
(**Hint:** Recycle the calculations from the previous part)

Scratch working page